Project: RePlastix Innovations: Transforming Plastic Waste into Sustainable Solutions Salesforce

Sumbitted By:

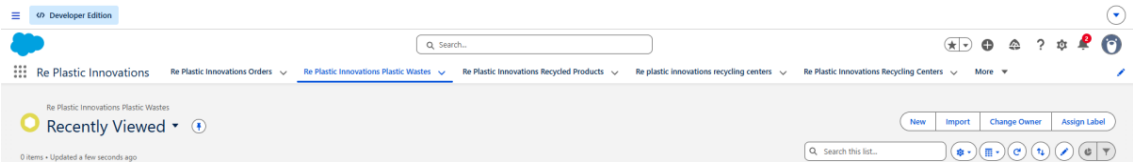
Project Overview

The Re-Plastic Innovation Project is a forward-thinking initiative aimed at transforming plastic waste from an environmental challenge into a valuable resource. By integrating circular economy principles, the project focuses on reducing plastic pollution through innovative recycling technologies, sustainable product design, and community-driven waste management systems.

Our approach begins at the source — reimagining how plastics are produced, used, and recovered. We leverage advanced sorting, chemical recycling, and upcycling methods to convert discarded plastics into high-quality raw materials for manufacturing. This not only diverts waste from landfills and oceans but also reduces the demand for virgin plastic production, lowering carbon emissions.

Beyond technology, the project emphasizes education, policy advocacy, and local entrepreneurship to create a self-sustaining ecosystem. By empowering communities, fostering green jobs, and encouraging responsible consumption, the Re-Plastic Innovation Project aims to set a replicable model for sustainable development.

Ultimately, our vision is to close the loop on plastic use, ensuring that every piece of plastic is reused, repurposed, or recycled — contributing to cleaner environments, stronger economies, and a healthier planet for future generations.



Objectives

To conceptualize, design, and implement innovative, sustainable, and economically viable solutions for the recycling, reuse, and reduction of plastic waste. This project aims to address the growing environmental challenges posed by plastic pollution by integrating advanced technologies, creative design thinking, and community-driven initiatives. Through the development of new methods for transforming discarded plastics into functional, high-value products, the project seeks to promote a circular economy, reduce landfill dependency, and lower carbon emissions. Additionally, it aspires to raise public awareness, encourage behavioral change, and foster collaboration among stakeholders—including local communities, industries, and policymakers—to create a scalable and replicable model for responsible plastic management that benefits both the environment and society.

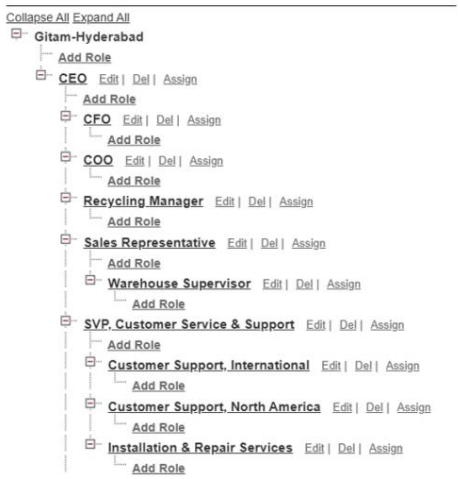
Phase 1: Requirement Analysis & Planning

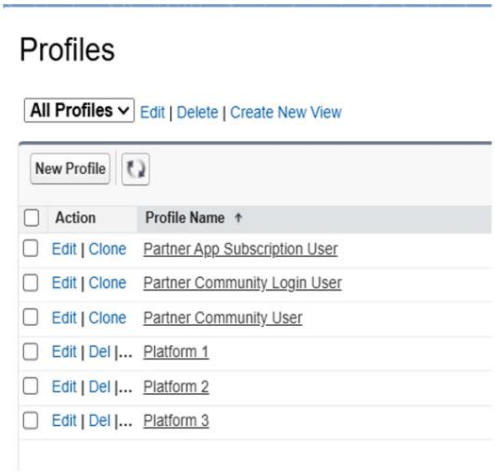
Business Requirements:

- **Centralized CRM for tracking waste, inventory, restock, and orders.**
- **Departmental access controls and real-time notifications.**

Scope:

- **Five custom objects: Plastic Waste, Recycling Center, Recycled Product, Order, Restock Request.**
- **Automated restock tasks when stock falls below thresholds.**
- **Validation rules to prevent incorrect entries.**
- **Role-based data access and sharing rules.**





Phase 2: Salesforce Development

Environment Setup:

- Developed in Salesforce Developer Edition.
- Lightning App “RePlastix Innovations” created with navigation tabs.
- Modular approach for Apex testing and debugging.

Custom Objects & Fields:

- Plastic Waste, Recycling Center, Recycled Product, Order, Restock Request.
- Key fields: Weight, Threshold, Stock Level, Status, Collection Date.

Validation Rules:

- Prevent future collection dates.
- Ensure order quantity > 0.

Automation (Flow Builder):

- Scheduled daily flow to check stock levels.
- Tasks auto-created when stock falls below threshold.
- Email notifications sent on restock approval.

Apex Classes & Triggers:

- InventoryManager: Handles stock deduction and restock approval.
- UpdateStockAfterOrder: Trigger after order insertion.
- UpdateStockAfterRestockApproval: Trigger after restock approval.
- EmailNotificationHelper: Sends email alerts.

Phase 3: UI/UX Development

A UI/UX project focused on re-plastic innovation aims to digitize the recycling ecosystem, bridging the gap between plastic waste collectors, processors, and consumers through sustainable design. Key initiatives include creating mobile applications for tracking plastic footprints, optimizing the collection process, and designing user-friendly interfaces for recycling marketplaces.

Project Core Content & Objectives

- Waste Tracking & Management: Developing systems that monitor plastic from source to landfill, facilitating a circular economy.
- Digital Marketplace (App): Building platforms like "Loak Yuk" to connect collectors (Tukang Loak) with users to increase the efficiency of recycling logistics.

- **Behavioral Change & Education: Using UI design to encourage sustainable habits, such as rewarding users for proper waste disposal.**
- **Upcycling Platform: Promoting the reuse of hard-to-recycle materials (PVC, PP) by showcasing products made from plastic waste.**

UI/UX Design Approach

- **Design Thinking Framework: Empathy-driven design to understand the needs of residents, waste collectors, and industries.**
- **Gamification & Rewards: Utilizing gamified digital wallets to motivate users to scan and deposit plastic items for rewards.**
- **Prototyping & Testing: Using tools like SUS (System Usability Scale) to validate that digital interfaces for plastic collection are intuitive and efficient.**
- **Environmentally Friendly UI: Adopting minimalist, energy-efficient, and "green-themed" interfaces that reflect sustainability**

Key Focus Areas

- **Plastic Sorting Applications: Mobile tools that help users identify and sort plastic waste, decreasing contaminants in the recycling stream.**
- **Sustainable Packaging Design: Designing interfaces that encourage the adoption of re-useable plastic alternatives or biodegradable alternatives.**
- **AI Integration: Utilizing AI/DB interfaces to analyze data for optimizing the mix of virgin and recycled resin, improving the quality of recycled plastic products.**

Phase 4: Data Migration, Testing & Security

A comprehensive project focused on data migration, testing, and security within the Re-Plastic Innovation sector involves transitioning legacy plastic recycling and management systems into modern, secure, and data-driven infrastructures. This ensures that innovations—such as chemical recycling, biological degradation, and AI-enabled sorting—are supported by reliable data and secure processes.

1. Project Content & Data Migration Strategy

The core of this project involves migrating data from legacy recycling systems to advanced platforms (e.g., cloud-based or AI-integrated ERPs) to support circular economy initiatives.

- **Data Scope: Inventory, sort, and transfer data related to plastic waste collection, sorting, polymer types, chemical recycling processing parameters, and AI-based sorting quality assurance metrics.**
- **Migration Approach:**

    - **Phased (Trickle) Approach: Recommended for continuous recycling operations to minimize downtime by moving smaller segments over time.**
    - **Data Cleaning: Eliminate "garbage in, garbage out" scenarios by cleaning legacy data (e.g., removing obsolete sorting logs) before migration.**
- **Mapping & Transformation: Develop detailed maps (ETL) to ensure that source field formats match the new database schema, particularly for tracking new materials.**

2. Testing Strategies on Re-Plastic Innovation

Testing is crucial to ensure that new technologies (e.g., AI-sorting algorithms) accurately classify plastics and that the system maintains data integrity.

- **Pre-Migration Testing: Validate source data quality, perform data profiling, and test data mapping to ensure compatibility.**
- **Migration Testing: Perform "data validation testing" to ensure that 100% of the data has been transferred without loss, using automated tools for verification.**
- **Post-Migration Testing (Data Reconciliation):**
    - **Reconciliation Tests: Compare source and target data counts, checksums, and key material totals.**
    - **Functional Testing: Ensure that the new system properly manages the chemical/biological degradation process data.**
- **Performance Testing: Simulate high-volume data ingestion to ensure the system supports real-time AI sorting and tracking.**

3. Security Considerations

Data security must be embedded throughout the project to protect proprietary recycling formulas and sensitive operational data.

- **Secure Transition: Use encrypted connections (HTTPS/TLS) during data transit.**
- **Data Access Control: Implement strict role-based access control (RBAC), ensuring only authorized personnel have access to critical process parameters or customer data.**
- **Data Masking/Anonymization: Mask sensitive data, especially if using third-party AI or cloud services for process optimization.**
- **Backup & Rollback Plan: Maintain complete, verified backups of the source system. Create a step-by-step rollback strategy to revert to the old system if the migration fails.**

- **Compliance: Adhere to environmental data reporting regulations and cybersecurity standards (e.g., ISO 27001).**

Phase 5: Deployment & Maintenance

Deployment Strategy:

- **Manual migration for demo org.**
- **Real-world deployment via Change Sets, Salesforce CLI, and Git.**

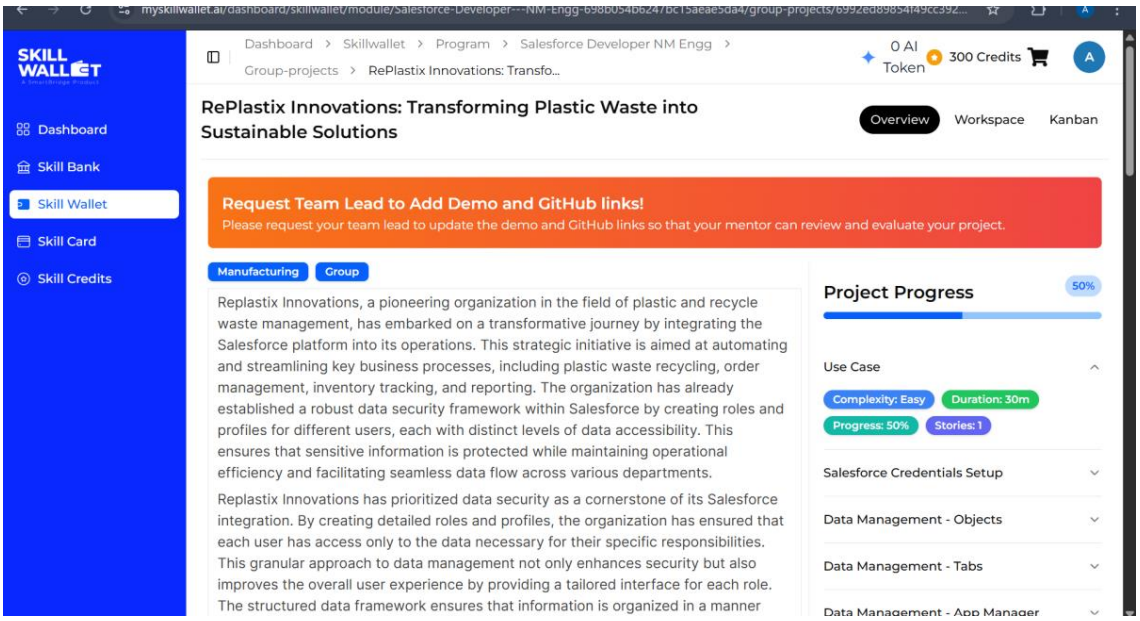
System Maintenance:

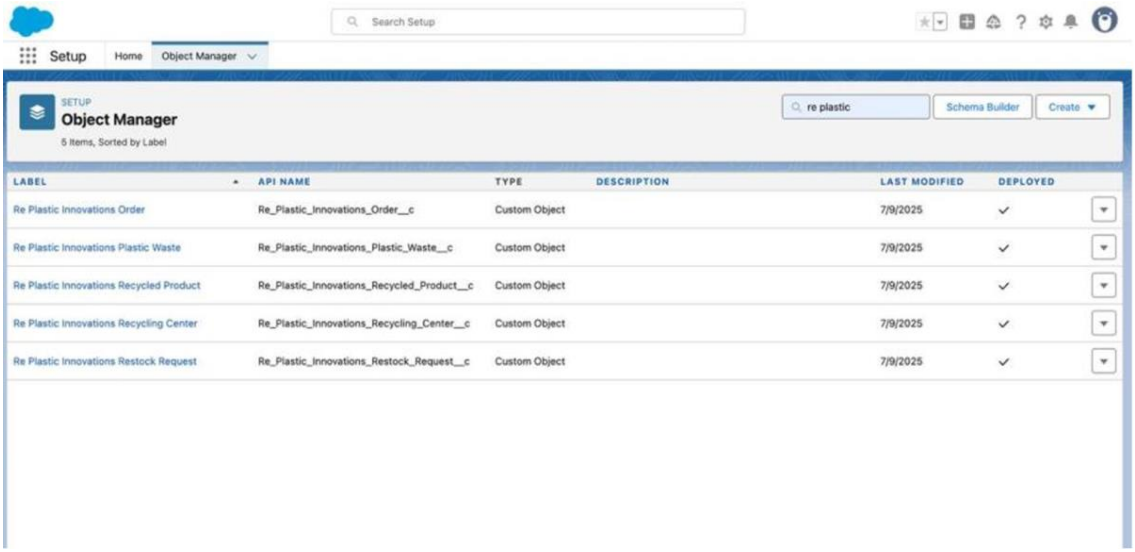
- **Scheduled flows monitored via Flow UI.**
- **Debug logs and exception logs for troubleshooting.**
- **Field history tracking for audit trails.**
- **Regular reviews of permissions and sharing rules.**

Troubleshooting Guide:

⍰ Apex errors: Debug logs.

- **Validation errors: Object Manager → Validation Rules.**
- **Flow failures: Flow Interview Logs.**
- **Permission issues: Profiles and role hierarchy.**
- **Data loading errors: Data Loader error files.**

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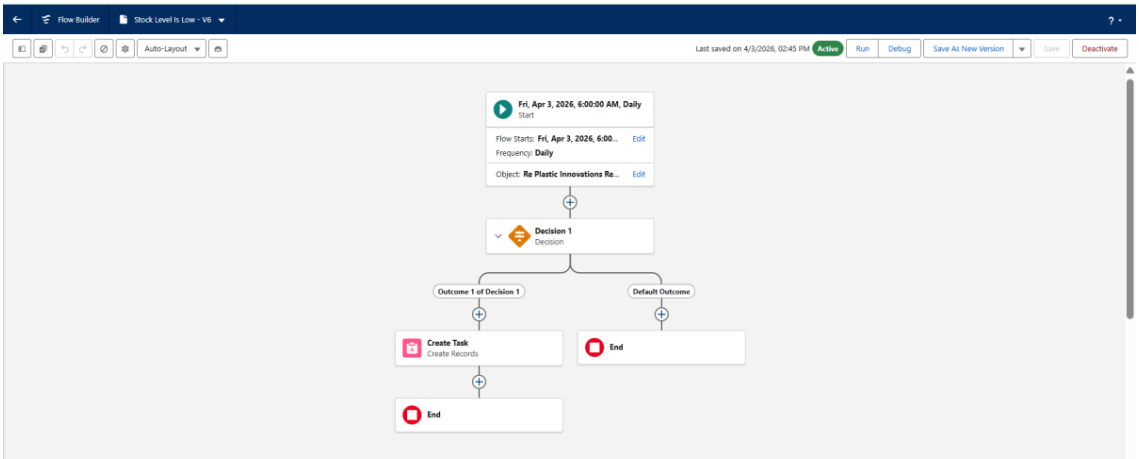
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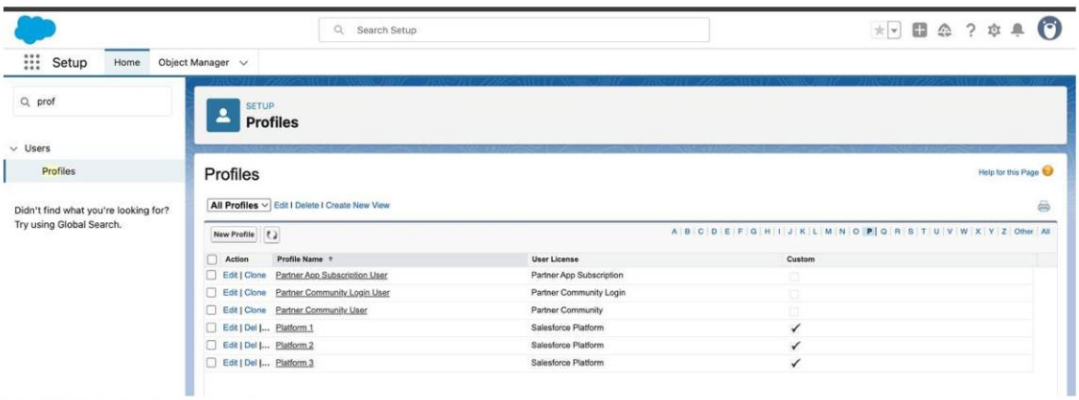
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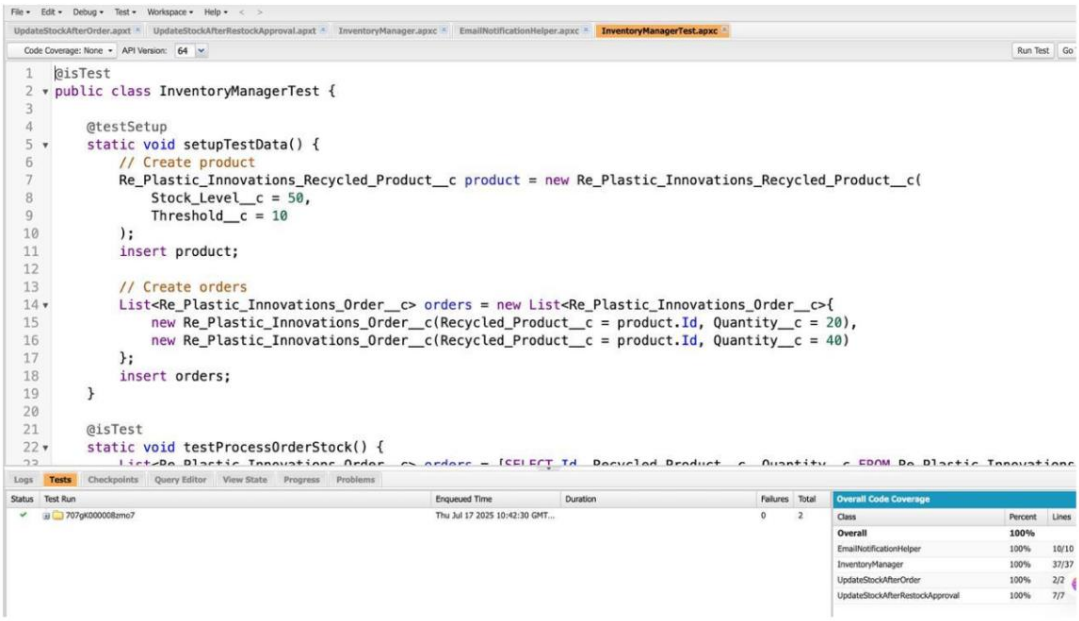
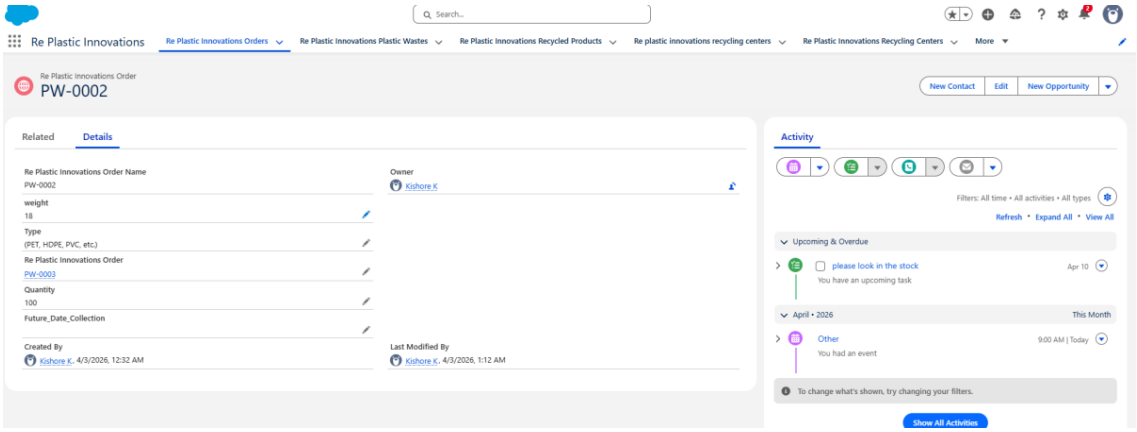


```
@isTest
public class InventoryManagerTest {

    @testSetup
    static void setupTestData() {
        // Create product
        Re_Plastic_Innovations_Recycled_Product__c product = new Re_Plastic_Innovations_Recycled_Product__c(
            Stock_Level__c = 50,
            Threshold__c = 10
        );
        insert product;

        // Create orders
        List<Re_Plastic_Innovations_Order__c> orders = new List<Re_Plastic_Innovations_Order__c>{
            new Re_Plastic_Innovations_Order__c(Recycled_Product__c = product.Id, Quantity__c = 20),
            new Re_Plastic_Innovations_Order__c(Recycled_Product__c = product.Id, Quantity__c = 40)
        };
        insert orders;
    }

    @isTest
    static void testProcessOrderStock() {
```



Conclusion

The RePlastix Innovations Salesforce CRM project demonstrates how technology can automate and streamline complex recycling operations. By integrating custom objects, validation rules, flows, Apex logic, and secure access models, the system addresses challenges like inventory shortages, delayed restocking, and inefficient order management.

This solution enhances productivity, ensures data accuracy, and supports sustainability goals through efficient resource management. It provides a scalable, user-friendly, and secure CRM tailored to RePlastix Innovations, laying a strong foundation for future enhancements.

Resources:

- **Trailhead Profile: Salesforce Trailblazer (salesforce.com in Bing)**

- **Demo Video: Google Drive Link (drive.google.com in Bing)**
- **GitHub Repository: RePlastix Innovations (github.com in Bing)**
- **https://drive.google.com/drive/folders/1X6_gkyt5Ba22hrfkCv-JMxXugV6lCfVB**